

**A Capstone Project Report on
Surface Water and Drinking Water Contaminants**

CS 896 Data Science Capstone

**By
Kavya Sri Modepu
Bindu Nagasai**

**Under the guidance of
Ast. Prof. Dr. Lokesh Das**

Wichita State University

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Abstract

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The main goal was to understand how contaminant levels behave across both datasets and to explore whether surface water conditions could be related to drinking water contamination. This project focuses on analyzing contaminant patterns in drinking water systems across Kansas and examining how these patterns relate to surface water conditions.

Two datasets were used in this study a municipal drinking water dataset and a surface water dataset obtained from the Water Quality Portal (WQP). The drinking water dataset required multiple cleaning steps, including unit conversion, handling non-numeric values, and creating variables such as standardized concentration values, exceedance indicators, and outlier flags.

The analysis showed that Barium, Fluoride, Selenium, Nitrate, and Chromium were the most frequently reported contaminants. Among these, nitrate showed significantly higher concentration levels compared to others, increasing from approximately 2800 ppb in 2019 to nearly 5000 ppb in 2024. Regression analysis indicated that year alone does not explain variation when all contaminants are combined, which suggests that analyzing contaminants separately gives more meaningful results.

Since the surface water dataset included latitude and longitude information, spatial analysis was possible for monitoring locations. Monitoring locations were mapped, and nitrate was analyzed across locations and years. The results showed that nitrate concentrations vary across sites and fluctuate over time without a clear linear trend. Site-level results also showed that higher values are distributed across multiple locations rather than concentrated in a single area.

A key limitation of this study is the absence of geographic identifiers in the drinking water dataset, which prevents direct linkage between surface water sources and drinking water systems. Therefore, the findings are presented as two related analyses rather than a direct causal relationship.

1. Introduction

In this project, we analyzed water quality data in Kansas to understand how contaminants behave in drinking water and surface water systems. The goal was to check whether contaminant patterns in surface water could also be seen in drinking water data.

At the beginning, we expected to directly connect both datasets using location information. However, during the analysis, we noticed that the drinking water dataset does not include latitude or longitude. Because of this, it was not possible to directly link the two datasets.

Due to this limitation, we decided to approach the project in two parts. First, we analyzed contaminant patterns in the drinking water dataset. Then, we performed spatial and temporal analysis on the surface water dataset separately.

Even though a direct connection was not possible, both datasets still showed useful contaminant patterns. During the analysis, nitrate behaved differently compared to most other contaminants, so we focused on it in more detail.

2. Problem Statement and Objectives

The main goal of this project was to analyze contaminant levels in drinking water systems across Kansas and compare them with patterns observed in surface water data. We wanted to understand which contaminants appear frequently, how their values change over time, and whether surface water analysis provides additional context for drinking water contamination.

The project focused on the following objectives:

- Identify the most frequently reported contaminants in the drinking water dataset and analyze their concentration levels.
- Examine trends in contaminant concentrations over time to understand whether values are increasing, decreasing, or remaining stable.
- Evaluate exceedance patterns by comparing observed values with regulatory limits (MCL) to identify potential risk areas.
- Analyze variability in contaminant data by identifying outliers and understanding extreme values.
- Perform spatial analysis of surface water data to understand how measurements vary across different monitoring locations.
- Focus on nitrate as a key contaminant and analyze its behavior in both drinking water and surface water datasets.
- Check whether the available data could be used to compare drinking water and surface water contaminant patterns.

During the analysis, it was found that the drinking water dataset does not include geographic identifiers such as latitude, longitude, or a shared location key. This limitation made it impossible to directly connect the two datasets.

During the analysis, we noticed that the drinking water dataset does not contain latitude, longitude, or another shared location field. Because of this, it was not possible to directly connect both datasets. Instead, we analyzed drinking water and surface water separately and compared the overall contaminant patterns.

3. Data Sources and Description

This project used two datasets: a drinking water dataset and a surface water dataset collected from the Water Quality Portal (WQP). Both datasets were used to analyze contaminant patterns in Kansas.

3.1 Drinking Water

The drinking water dataset contains contaminant measurements collected from municipal water systems. The key variables used in this dataset include

- Contaminant name
- Reported concentration value (Highest Value)
- Measurement unit (ppm, mg/L, $\mu\text{g/L}$)
- Maximum Contaminant Level (MCL)
- Collection date and year
- Source or system identifier

One challenge in this dataset was that contaminant values were recorded using different units such as ppm, mg/L, and $\mu\text{g/L}$. Some records also contained values like "<0.5" and "ND", which could not be used directly in calculations. These records had to be cleaned and standardized before analysis.

After preprocessing, all concentration values were converted into a common unit (ppb), and only valid numeric records were retained. Duplicate entries were removed, and the dataset was prepared for further analysis.

3.2 Surface Water Dataset The surface water dataset was obtained from the Water Quality Portal (WQP), which provides water quality measurements collected from monitoring locations across the United States.

The key variables used in this dataset include

- Monitoring Location Identifier
- Monitoring Location Name
- Date of measurement
- Characteristic name (e.g., nitrate, pH, turbidity)

- Measurement value
- Unit
- Latitude and longitude

Unlike the drinking water dataset, this dataset includes geographic coordinates, which allows for spatial analysis. Because latitude and longitude were available, the monitoring locations could be mapped and analyzed spatially.

After cleaning, the surface water dataset contained approximately 17,000 records across 164 monitoring locations. Records with missing geographic or measurement values were removed to ensure reliable analysis.

3.3 Focus on Nitrate

During initial analysis, nitrate was identified as one of the most important contaminants because its concentration values were much higher than most other contaminants. Because of this, nitrate was selected for detailed analysis across both datasets.

In the surface water dataset, nitrate records were filtered to create a subset of approximately 420 observations across 74 monitoring locations. This subset was used for spatial and temporal analysis.

3.4 Data Limitations A key limitation identified in this project is the lack of geographic information in the drinking water dataset. Specifically, the dataset does not include latitude, longitude, or a shared identifier that can be used to link it with surface water monitoring locations.

Because of this, it is not possible to directly connect drinking water systems with specific surface water sources. This limitation affects the ability to perform combined spatial analysis across the two datasets.

During this stage, we realized that combining both datasets directly would be difficult due to missing geographic information.

4. Data Preparation

Before analysis, both datasets needed multiple cleaning and preprocessing steps because the data came from different sources and contained inconsistent formats.

4.1 Drinking Water Data Preparation

The drinking water dataset needed several cleaning steps due to variations in how contaminant values were recorded. One of the main issues was the use of different measurement

units such as ppm, mg/L, and $\mu\text{g/L}$. To make the data comparable across all contaminants, all values were converted into a common unit, parts per billion (ppb).

Another challenge was handling non-numeric entries such as “<0.5” and “ND” (non-detect). These values were processed by extracting the numeric component where possible or treating them as missing values.

Duplicate records were removed to avoid repeated observations. In addition, rows with missing or invalid values in key fields such as contaminant name, concentration, and year were excluded.

To support analysis, new variables were created. The standardized concentration value (Highest_ppb) was used as the primary variable for analysis. Regulatory limits were also converted into ppb (MCL_ppb) to allow direct comparison with observed values. An exceedance indicator (Exceed_MCL) was created to identify cases where contaminant levels exceeded regulatory limits.

An outlier detection step was also performed using the interquartile range (IQR) method for each contaminant.

4.2 Surface Water Data Preparation

For the surface water dataset, the cleaning process mainly focused on preparing the data for mapping and trend analysis. Relevant columns such as measurement value, characteristic name, date, latitude, and longitude were selected for analysis.

Measurement values and geographic coordinates were converted into numeric format to ensure consistency. Records with missing latitude or longitude values were removed, as they cannot be used for mapping or spatial analysis.

Date values were converted into a standard format, and a year variable was extracted to support trend analysis over time. The dataset contained multiple water quality characteristics, but nitrate was selected for detailed analysis because it showed higher values in the drinking water dataset. A filtered subset of nitrate observations was created, which was used for spatial visualization and temporal analysis.

5. Methodology

During the cleaning phase, we faced issues with inconsistent units such as ppm, mg/L, and ppb. To make the data comparable, we converted all values into ppb. We also had to handle values like “<0.5” and “ND”, which had to be cleaned manually before analysis.

After cleaning, we performed exploratory analysis to understand which contaminants appear frequently and how their values are distributed.

We then tried to analyze trends over time and applied a regression model to check whether year could explain changes in concentration. However, we observed that combining all contaminants in a single model did not produce meaningful results.

For the surface water dataset, we used geographic information to visualize monitoring locations and analyze spatial patterns. We focused on nitrate for deeper analysis since it showed higher values in the drinking water dataset.

5.1 Drinking Water Analysis Approach

During the cleaning process, we noticed that the dataset had multiple units such as ppm, mg/L, and ppb. To make the data consistent, we converted all values into ppb. We also handled values like "<0.5" and "ND", which had to be cleaned before analysis.

After cleaning, we explored the data to identify the most frequently occurring contaminants. We then analyzed trends over time and applied a regression model to check whether year could explain changes in concentration. However, we found that combining all contaminants in one model did not show a clear relationship.

5.2 Surface Water Analysis Approach

For the surface water dataset, we used the available geographic information to understand spatial patterns. We cleaned the dataset by converting values into numeric format and removing records with missing latitude and longitude. After cleaning, the monitoring locations were mapped using latitude and longitude values.

We focused on nitrate for detailed analysis since it showed higher values in the drinking water dataset. We analyzed how nitrate values vary across locations and over time. From this, we observed that nitrate values fluctuate and are spread across multiple locations rather than being concentrated in one specific area.

5.3 Combined Approach

During the project, we initially expected to directly connect the drinking water and surface water datasets. However, the drinking water dataset did not contain geographic identifiers such as latitude or longitude. Because of this, both datasets were analyzed separately while keeping the focus on nitrate and other major contaminants.

6. Drinking Water Analysis

6.1 Contaminant Frequency Analysis

The first step in the analysis was to identify the most frequently reported contaminants in the dataset. The results showed that Barium, Fluoride, Selenium, Nitrate, and Chromium appeared much more often than the other contaminants.

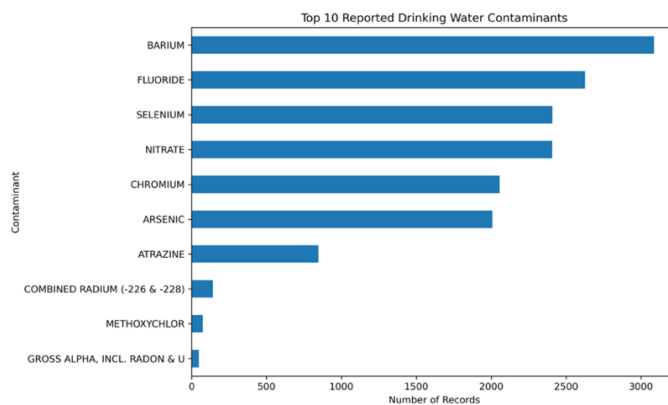


Figure 1. Top 10 Reported Drinking Water Contaminants

6.2 Trend Analysis of Top Contaminants

After identifying the most common contaminants, their concentration trends were analyzed across different years. The results show that most contaminants, including Barium, Fluoride, Selenium, and Chromium, remain relatively stable over time without major changes. In contrast, nitrate behaves differently compared to the other contaminants, with much higher concentration levels compared to other contaminants.

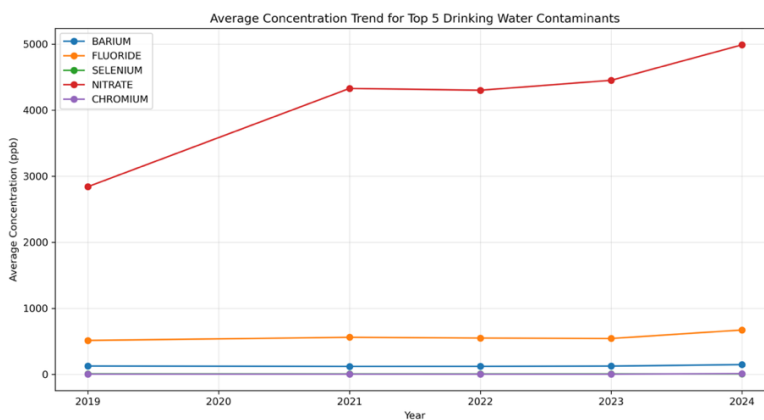


Figure 2. Average Concentration Trends for Top 5 Contaminants

6.3 Nitrate Analysis Nitrate was analyzed separately because it showed the highest concentration levels among all contaminants. The trend analysis indicates that nitrate values increase from approximately 2800 ppb in 2019 to nearly 5000 ppb in 2024, with some fluctuations between years.

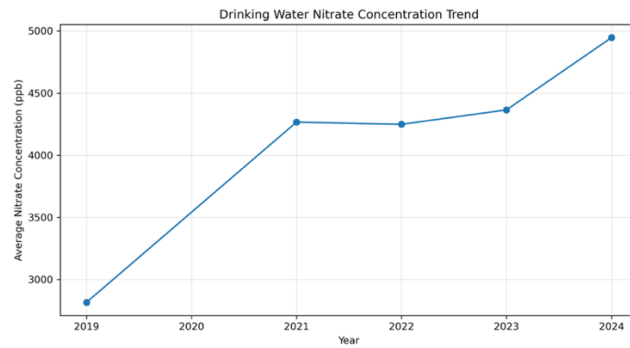


Figure 3. Drinking Water Nitrate Concentration Trend (2019–2024)

From the graph, nitrate behaves differently from the other contaminants and shows larger changes over time.

6.4 Regression Analysis

A regression model was used to check whether year could explain changes in contaminant concentration. The results show a very low R-squared value and a high p-value, indicating that there is no significant linear relationship.

The results showed that combining all contaminants into one model was not useful because each contaminant behaves differently.

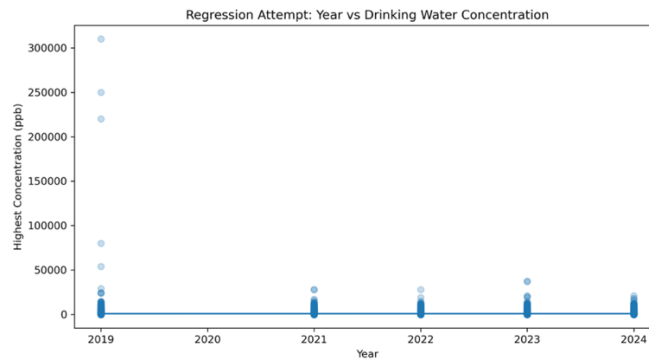


Figure 4. Regression Analysis (Year vs Concentration)

6.5 Exceedance Analysis

Exceedance analysis was conducted by comparing observed contaminant values with regulatory limits (MCL).

The results showed that some contaminants had high exceedance rates but only a small number of observations. Nitrate had a moderate exceedance rate but appeared in a much larger number of records.

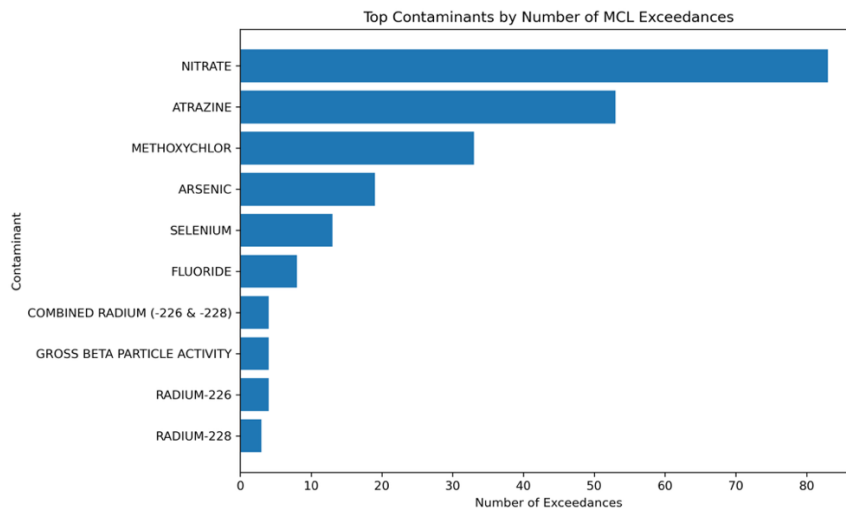


Figure 5. Contaminants by Number of MCL Exceedances

7. Surface Water Analysis

7.1 Monitoring Location Distribution

The first step in the surface water analysis was to visualize the distribution of monitoring locations.

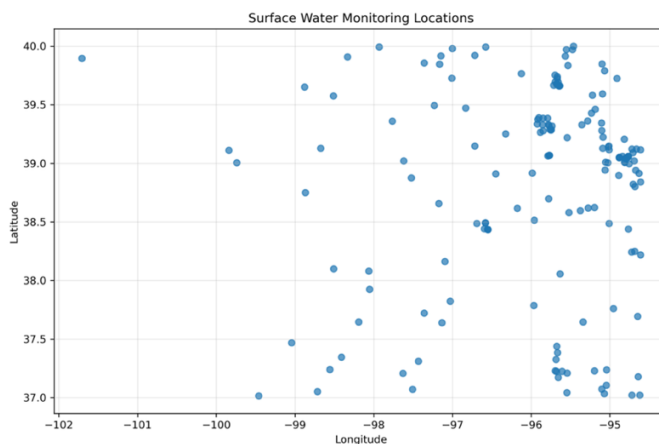


Figure 6. Distribution of Surface Water Monitoring Locations

Some regions have a higher concentration of monitoring sites, while others have fewer. The plot shows that monitoring locations are spread across Kansas, but the distribution is uneven. Some areas contain more monitoring sites than others.

7.2 Surface Water Characteristics

The dataset includes several surface water characteristics such as turbidity, pH, dissolved oxygen, temperature, and nitrate. Among these, nitrate was selected for detailed analysis because it showed higher values in the drinking water analysis.

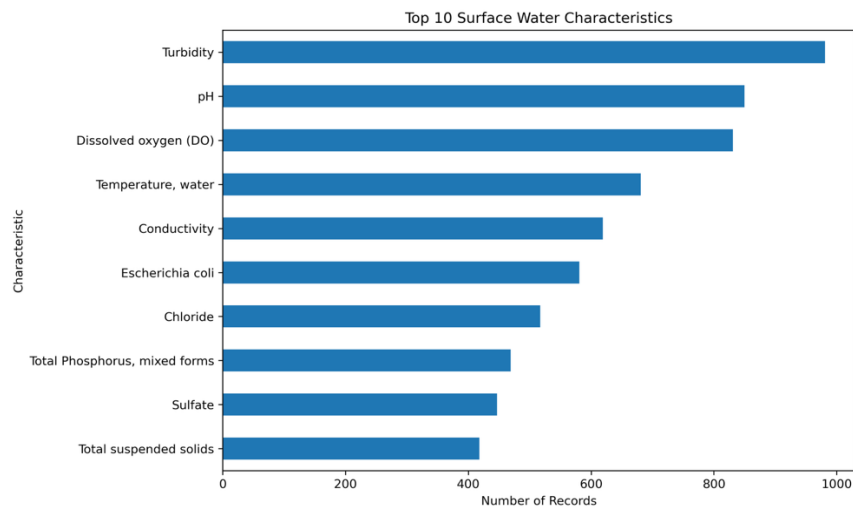


Figure 7. Most Frequently Measured Surface Water Characteristics

7.3 Nitrate Spatial Analysis

Nitrate records were filtered from the surface water dataset for spatial analysis. The filtered dataset contains approximately 420 nitrate measurements across 74 monitoring locations. The results showed that higher nitrate values are spread across multiple locations instead of one specific hotspot.

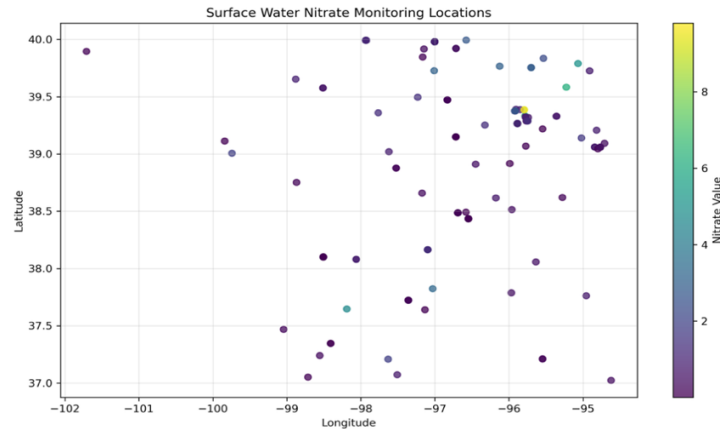


Figure 9. Spatial Distribution of Nitrate in Surface Water

7.4 Nitrate Temporal Analysis

Yearly average nitrate values were analyzed to observe how concentrations changed over time. The results fluctuate across years and do not show a clear increasing or decreasing trend.

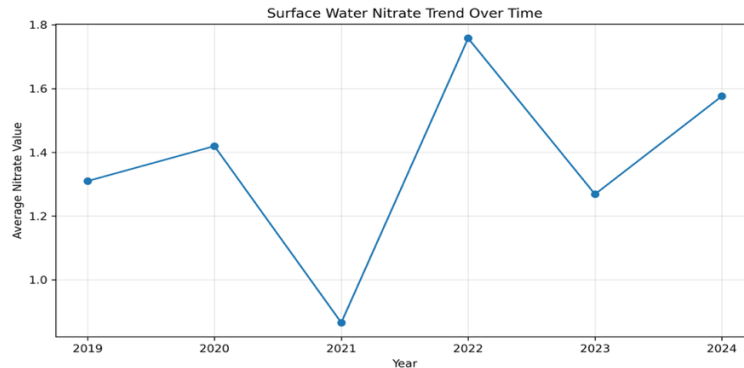


Figure 10. Surface Water Nitrate Trend Over Time

7.5 Site-Level Nitrate Analysis

A site-level summary was created to examine how nitrate values differ across monitoring locations. The results indicate that some locations have higher average nitrate values than others. Higher nitrate values appeared across multiple monitoring locations rather than one specific area.

Table 1. Summary of Nitrate Values by Monitoring Location

MonitoringLocationID	Records	Mean_Value	Median_Value	Max_Value	Latitude	Longitude
PRAIRIEBAND-MW11	2	8.8	8.8	9.8	39.3858	-95.7932
PRAIRIEBAND-MW08	2	6.7	6.7	8.6	39.3318	-95.7814
NARS_WQX-NRS_KS-10435	1	6.19	6.19	6.19	39.58314228	-95.22446835
NARS_WQX-NRS_KS-10449	1	5.02	5.02	5.02	39.78982916	-95.06722268
NARS_WQX-NRS_KS-10249	1	4.65	4.65	4.65	37.64636548	-98.19185293
NARS_WQX-NRS_KS-10248	1	3.3074	3.3074	3.3074	37.82311746	-97.0303315
NARS_WQX-NRS_KS-10210	2	3.1957	3.1957	3.6902	39.75447276	-95.69903262
NARS_WQX-NRS_KS-10368	1	3.1	3.1	3.1	39.72738834	-97.01089354
NARS_WQX-NRS_KS-10404	1	3.01	3.01	3.01	39.76648562	-96.12754016
NARS_WQX-NRS_KS-10205	1	2.7479	2.7479	2.7479	39.99441929	-96.57905059
NARS_WQX-NRS_KS-10443	1	2.26	2.26	2.26	39.83518293	-95.53318623
NARS_WQX-NRS_KS-10468	1	1.84	1.84	1.84	39.00631846	-99.74227905
NARS_WQX-NRS_KS-10367	1	1.84	1.84	1.84	37.20753864	-97.63141656
21NEB001_WQX-SLB1LBLUE000	155	1.639869677	1.73	7.55	39.980149	-97.003893
NARS_WQX-NWC_KS-10130	1	1.63	1.63	1.63	39.139578	-95.020855
21NEB001_WQX-SRE1REPUB000	87	1.486218391	1.25	3.74	39.992569	-97.931967
NARS_WQX-NRS_KS-10050	1	1.3473	1.3473	1.3473	39.49522569	-97.23190219
NARS_WQX-NRS_KS-10046	1	1.3143	1.3143	1.3143	39.25252179	-96.32659742
NARS_WQX-NRS_KS-10360	1	1.27	1.27	1.27	39.35926743	-97.76596877
PRAIRIEBAND-SCO7	18	1.151111111	1.05	3.3	39.2658	-95.8864
PRAIRIEBAND-SCO3	25	1.1228	0.83	2.9	39.3758	-95.9183
PRAIRIEBAND-LSCO6	10	1.107	1.15	1.8	39.2888	-95.7483
PRAIRIEBAND-MW10	1	1.1	1.1	1.1	39.3873	-95.8494
NARS_WQX-NRS_KS-10269	2	1.03485	1.03485	2.0217	39.92164243	-96.71631982
PRAIRIEBAND-MW09	1	0.97	0.97	0.97	39.3904	-95.9053
NARS_WQX-NRS_KS-10255	1	0.9048	0.9048	0.9048	39.65317317	-98.88144118
NARS_WQX-NRS_KS-10226	2	0.83935	0.83935	1.1567	39.32949678	-95.35688326
NARS_WQX-NRS_KS-10208	1	0.8141	0.8141	0.8141	39.01991425	-97.62109954
NARS_WQX-NRS_KS-10222	1	0.8079	0.8079	0.8079	39.20750693	-94.81716042
NARS_WQX-NRS_KS-10039	1	0.7685	0.7685	0.7685	39.72624414	-94.91122091
NARS_WQX-NRS_KS-10431	1	0.724	0.724	0.724	38.6157999	-96.17854363
NARS_WQX-NRS_KS-10246	1	0.7193	0.7193	0.7193	37.76116475	-94.9562043
NARS_WQX-NRS_KS-10199	1	0.6977	0.6977	0.6977	37.07185357	-97.50849573
EPA_R7_WQX-RSTIP09	3	0.695666667	0.27	1.55	38.163689	-97.097927
NARS_WQX-NRS_KS-10051	2	0.6841	0.6841	0.7043	39.06027253	-94.84194972
PRAIRIEBAND-LSCO3	8	0.6675	0.665	1.2	39.3225	-95.7758
NARS_WQX-NRS_KS-10196	2	0.6445	0.6445	0.74	39.57629487	-98.51414655
PRAIRIEBAND-MW03	1	0.61	0.61	0.61	39.318614	-95.738642

MonitoringLocationID	Records	Mean_Value	Median_Value	Max_Value	Latitude	Longitude
PRAIRIEBAND-LSCO4	4	0.57	0.6	0.63	39.2887	-95.7617
NARS_WQX-NRS_KS-10264	1	0.5171	0.5171	0.5171	38.91041431	-96.45214179
NARS_WQX-NWC_KS-10102	2	0.5151	0.5151	0.991	38.07983	-98.06508
NARS_WQX-NRS_KS-10250	1	0.5087	0.5087	0.5087	38.05708438	-95.63541799
EPA_R7_WQX-KVC5	1	0.491	0.491	0.491	39.05908628	-94.76684306
NARS_WQX-NWC_KS-10143	1	0.478	0.478	0.478	37.787233	-95.967877
NARS_WQX-NRS_KS-10293	1	0.4029	0.4029	0.4029	38.51416207	-95.96179912
EPA_R7_WQX-CON2	1	0.374	0.374	0.374	39.093076	-94.71003265
NARS_WQX-NRS_KS-10289	1	0.3634	0.3634	0.3634	37.02237885	-94.6211974
EPA_R7_WQX-RSTIP10	4	0.3575	0.35	0.71	37.722646	-97.358462
NARS_WQX-NRS_KS-10252	1	0.332	0.332	0.332	38.65767114	-97.17331128
NARS_WQX-NWC_KS-10129	1	0.331	0.331	0.331	37.239256	-98.557535
NARS_WQX-NRS_KS-10361	1	0.33	0.33	0.33	39.91711927	-97.14841713
EPA_R7_WQX-RSTIP03	5	0.25282	0.132	0.768	39.472247	-96.832223
NARS_WQX-NRS_KS-10448	1	0.215	0.215	0.215	37.4680573	-99.04453627
EPA_R7_WQX-KVC4	1	0.196	0.196	0.196	39.05770136	-94.76781834
NARS_WQX-NRS_KS-10245	1	0.189	0.189	0.189	38.75125952	-98.87007228
NARS_WQX-NRS_KS-10262	1	0.1881	0.1881	0.1881	38.61995291	-95.27727329
NARS_WQX-NRS_KS-10093	2	0.1469	0.1469	0.1822	37.20970471	-95.54539085
EPA_R7_WQX-RSTIP04	2	0.1435	0.1435	0.153	38.877043	-97.52317
EPA_R7_WQX-KVC6	1	0.139	0.139	0.139	39.060303	-94.7665188
EPA_R7_WQX-RSTIP08	5	0.0956	0.131	0.154	38.486057	-96.691025
NARS_WQX-NRS_KS-10446	1	0.079	0.079	0.079	39.84655733	-97.16511325
EPA_R7_WQX-KVC2	1	0.0752	0.0752	0.0752	39.0444992	-94.79866752
EPA_R7_WQX-RSTIP05	10	0.065	0.02	0.147	38.10035	-98.509398
EPA_R7_WQX-RSTIP02	5	0.0606	0.02	0.223	39.149088	-96.718446
EPA_R7_WQX-RSTIP07	10	0.0591	0.02	0.219	38.43461	-96.54928
NARS_WQX-NRS_KS-10290	1	0.0548	0.0548	0.0548	37.05079383	-98.71547714
NARS_WQX-NRS_KS-10235	1	0.051	0.051	0.051	39.89576845	-101.7077863
EPA_R7_WQX-LBB2	1	0.0442	0.0442	0.0442	39.0685004	-95.77104932
NARS_WQX-NRS_KS-10412	1	0.03	0.03	0.03	38.91634574	-95.9883252
NARS_WQX-NRS_KS-10345	1	0.01	0.01	0.01	39.21883246	-95.54474928
NARS_WQX-NRS_KS-10484	1	0.008	0.008	0.008	38.49167	-96.58222
NARS_WQX-NWC_KS-10098	2	0.0072	0.0072	0.0135	37.34562	-98.41065
NARS_WQX-NRS_KS-10271	1	0.0054	0.0054	0.0054	39.11269578	-99.84059319
NARS_WQX-NWC_KS-10142	1	0.0016	0.0016	0.0016	37.639357	-97.134782

8. Comparative Discussion

From the drinking water analysis, nitrate was identified as the main contaminant of interest because it showed much higher concentration levels compared to the other contaminants. The trend increased from approximately 2800 ppb in 2019 to nearly 5000 ppb in 2024. In comparison, contaminants such as Barium, Fluoride, Selenium, and Chromium did not show major changes over time.

In the surface water analysis, nitrate was also selected as the main contaminant analyzed. However, the behavior observed in surface water was different from drinking water. The results fluctuated across years and did not show a consistent trend.

The spatial analysis showed that nitrate values were spread across multiple monitoring locations. No single region consistently showed the highest nitrate values.

Both datasets showed nitrate as an important contaminant, but the results could not be directly compared because the datasets were structured differently.

A major limitation in this comparison was the absence of geographic information in the drinking water dataset. Since the dataset does not include latitude, longitude, or a shared location identifier, it was not possible to connect drinking water systems with nearby surface water monitoring locations. Because of this limitation, the project could not confirm a direct relationship between the two datasets.

Even though the datasets could not be directly connected, both analyses showed that nitrate appears consistently in drinking water and surface water data.

9. Limitations

During this project, we faced several limitations that affected the analysis.

- The drinking water dataset does not include geographic information such as latitude or longitude. Because of this, we were not able to directly connect drinking water systems with surface water monitoring locations.
- The drinking water and surface water datasets represent different types of measurements, which made direct comparison difficult.
- A significant amount of time was spent handling inconsistent units such as ppm, mg/L, and ppb, along with values like "<0.5" and "ND".
- Some records contained missing or incomplete values, which required cleaning and removal before analysis.

- The analysis mainly focused on nitrate and a few major contaminants instead of all contaminants in the dataset.
- External factors such as rainfall, seasonal changes, and land use were not included in the analysis.

Even with these limitations, the datasets still showed useful contaminant patterns.

10.Future Work and Recommendations

The following improvements could help extend this project in the future:

- Include geographic identifiers for drinking water systems so they can be connected with nearby surface water monitoring locations.
- Include additional environmental variables such as rainfall, land use, and seasonal factors.
- Expand the analysis to include more contaminants beyond nitrate.
- Apply more advanced modeling methods if more detailed data becomes available.
- Improve consistency between datasets to support better comparison and analysis.

More complete geographic and environmental data could help improve the connection between drinking water and surface water analysis.

11.Conclusion

In this project, we analyzed drinking water and surface water datasets to understand contaminant patterns in Kansas. The results showed that nitrate had much higher concentration levels compared to most other contaminants and changed more across different years.

Most of the other contaminants did not show major changes over time. The regression analysis also showed that combining all contaminants into one model did not produce a clear relationship.

In the surface water analysis, nitrate values varied across monitoring locations and changed across years, but no consistent trend was observed. Higher nitrate values were spread across multiple locations instead of one specific area.

One major limitation of this project was the lack of geographic information in the drinking water dataset. Because of this, it was not possible to directly connect drinking water systems with surface water monitoring locations.

Even though the datasets could not be directly linked, both analyses showed that nitrate appeared consistently in drinking water and surface water data.

12. References

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